

Howard University
 College of Engineering & Architecture
 Department of Electrical Engineering & Computer Science
EECE 306-01: Fundamentals of Electromagnetics Laboratory
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Lab 4: The Quadrature Engine and Continuous Charge Distributions

Objectives

By the end of this session you will have

- built the numerical integrator that every remaining lab runs on,
- validated the geometry engine without invoking any physics at all,
- measured the observed order of accuracy of four quadrature rules,
- found out for yourself how long a finite line has to be before it looks infinite.

1. The most reused module in the course

Eight functions are due today, which is the largest single build of the semester. It is worth the effort because of what runs through this module afterwards.

Every continuous charge distribution in Labs 5 through 8 is a call to `em.quad`. Every closed surface integral in the Gauss's law tests of Lab 5 and Lab 10 is a call to `em.quad.surf`. Every closed path integral in Lab 6 and Lab 10 is a call to `em.quad.line`. Every current distribution in Labs 9 and 11 is a call to `em.quad`. The flux calculation in Lab 12 is a call to `em.quad.surf`.

Write it once, write it well, and the rest of the semester is assembly.

2. Part I. Nodes and weights

2.1 `em.quad.nodes(a, b, N, rule)`

Returns N nodes and N weights on the interval $[a, b]$ for one of four rules, 'midpoint', 'trapz', 'simpson', 'gauss'. The default when `rule` is omitted is 'midpoint'.

Midpoint and trapezoid you can write from the definition. Simpson requires an odd number of nodes, so decide what your function does when handed an even N , and document it. For the Gauss rule, use the Golub-Welsch approach to obtain the Legendre nodes and weights, and map from the reference interval $[-1, 1]$ to $[a, b]$.

The contract is that for any smooth f ,

$$\int_a^b f(t) dt \approx \sum_{i=1}^N f(t_i) w_i$$

so a caller never needs to know which rule produced the nodes.

2.2 The three geometry wrappers

`em.quad.line`, `em.quad.surf` and `em.quad.vol` take a parametrization as a function handle and return element positions together with weights that already carry the Jacobian.

For a curve $\vec{r}(t)$,

$$w_i = \left| \frac{d\vec{r}}{dt} \right|_{t_i} w_i^{\text{node}}$$

For a surface $\vec{r}(u, v)$,

$$w_i = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|_i w_i^{\text{node}}$$

For a volume $\vec{r}(u, v, w)$ the weight uses the absolute value of the Jacobian determinant. Take the parameter derivatives by central differences with a step chosen relative to the parameter span, not as a fixed absolute number, since a parameter running from 0 to 2π and one running from 0 to 10^{-3} need different steps.

3. Part II. Testing the geometry before touching the physics

Here is the test that makes this lab safe, and it uses no field theory whatsoever.

If the weights carry the Jacobian correctly, then *summing the weights* returns the measure of the domain. Nothing else is required to be true for this to work.

```
circ = @(t) [cos(t), sin(t), zeros(size(t))];
[~, w] = em.quad.line(circ, [0 2*pi], 200);
em.test.assertClose(sum(w), 2*pi, 1e-6, 'unit circle arc length');
```

Required cases.

Domain	Parametrization	Sum of weights
Unit circle	$(\cos t, \sin t, 0), t \in [0, 2\pi]$	2π
Unit sphere surface	$(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$	4π
Unit ball	spherical with $r \in [0, 1]$	$4\pi/3$
Unit square	$(u, v, 0), u, v \in [0, 1]$	1

Debug the geometry separately from the physics

When an integral comes out wrong, there are two candidate culprits, the geometry and the integrand. Testing them together means guessing. Testing the geometry alone, by summing weights, tells you immediately which half is broken.

This is a professional habit worth forming now. Large simulation codes are built as layers precisely so that each layer can be tested without the ones above it. A student who checks the sphere area before computing a flux through the sphere will find the missing $\sin \theta$ in thirty seconds. A student who does not will spend an afternoon suspecting Coulomb's law.

4. Part III. Convergence, measured rather than assumed

Implement `em.test.convergence(errFcn, Nlist, opts)`. It evaluates `errFcn(N)` for each N in the list, fits a straight line to $\log(\text{error})$ against $\log(N)$, and returns the *negative* of the slope as the observed order p . It also draws the log-log plot.

Every convergence study for the rest of the course goes through this one function, which is why the plots will all look the same and be comparable.

Choose a definite integral whose exact value you know and study all four rules. Report a table of observed orders. Midpoint and trapezoid should come out near second order, and Simpson near fourth.

The Gauss rule will not fit a clean power law, and explaining why is a required part of your notebook. An N point Gauss rule is exact for polynomials up to degree $2N - 1$, so for a smooth integrand its error falls faster than any fixed power of N . Fitting a straight line to that data gives a slope, but the slope is not an order of accuracy in the same sense, and reporting it as one would be a misuse of the tool. Recognizing when a tool does not apply is part of using it.

Convergence order as a bug detector

The reason this matters is not curiosity about numerical analysis. An observed order that is lower than the theoretical order is one of the most reliable bug signals available. A second order method that measures as first order almost always has a boundary handled incorrectly, or an off-by-one in the node placement. You will use this diagnostic in anger in Lab 7 and again in Lab 8.

5. Part IV. Continuous charge distributions

Implement `em.src.lineCharge`, `em.src.surfCharge` and `em.src.volCharge`. Each builds the element list from the corresponding `em.quad` call and sets `q` from the density.

The density argument may be a scalar for a uniform distribution or a function handle evaluated at the element positions. Because the handle is evaluated at `pos`, a nonuniform distribution costs nothing beyond writing the handle.

Now notice what did *not* have to change. `em.field.E` from last week computes the field of these new sources without a single edit, because it never knew what kind of source it was looking at.

5.1 When does finite look infinite

Two studies, both of which run the same reasoning in different geometries.

Line. A uniform line charge ρ_L of length $2L$ centered on the z axis produces a field at a point on the y axis at distance d . The infinite line result is

$$E_{\text{inf}} = \frac{\rho_L}{2\pi\epsilon_0 d}$$

Plot the ratio of the finite result to the infinite result against L/d and identify the value of L/d at which the ratio reaches 0.99.

Sheet. A uniformly charged disk of radius a produces a field on its axis at height h . The infinite sheet result is

$$E_{\text{inf}} = \frac{\rho_S}{2\epsilon_0}$$

Plot the ratio against a/h and again find where it reaches 0.99.

Both are limit tests, and they check something no closed form check can, namely that your integrator behaves correctly as the domain grows and the integrand becomes slowly varying. State in your notebook what the two threshold aspect ratios came out to be, and comment on whether the two geometries approach their limits at similar rates.

Reference sheet

A one page reference for symbolic differentiation is distributed with this lab. It is optional and is provided for students who want to confirm one of these integrals analytically. Nothing in the graded work requires it.

Required tests

Write `tests/test_lab04.m` containing at least these checks.

1. Sum of weights equals 2π for the unit circle, 4π for the unit sphere surface, $4\pi/3$ for the unit ball, 1 for the unit square.
2. `em.quad.nodes` integrates a cubic exactly with the Simpson rule and with a two point Gauss rule.
3. Observed order near 2 for midpoint and trapezoid, near 4 for Simpson, on a smooth test integral.
4. A uniform line charge discretized into many elements reproduces the point charge field far away, to within one percent at a distance of twenty times its length.
5. The finite line ratio exceeds 0.99 at your reported L/d and falls below it just under that value.
6. The disk ratio exceeds 0.99 at your reported a/h .
7. A nonuniform line charge with density handle $\rho_L(z) = z$ produces zero net charge when integrated symmetrically about the origin, checked by summing `q .* w`.

Deliverables

Submit through Canvas by the start of the next lab session.

1. **Toolbox snapshot**, `TeamNN_Lab04.zip`.
2. **Lab notebook**, published script exported to PDF, three to five pages, with the table of observed orders, your explanation of the Gauss rule anomaly, and the two aspect ratio plots.
3. **Test output** pasted into the notebook.

Looking ahead

Next week Gauss's law stops being a theorem to admire and becomes an assertion your code either satisfies or fails. Read Sections 2.6 through 2.8 of Wentworth before you come.